



REVENUE ANALYTICS CASE STUDY

Google Merchandise Store: Diagnosing a Revenue Decline

A data-driven diagnosis of a conversion-driven revenue decline — with a recommended product intervention.

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THE PROBLEM

Revenue fell while traffic held steady

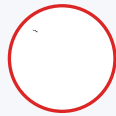
Sessions stayed flat across the quarter, so the drop had to be a conversion problem, a mix-shift problem, or something seasonal. Traffic alone couldn't explain it.



SESSIONS (3 MO.)

267,116

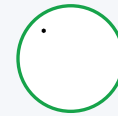
Held essentially flat, Nov – Jan



CONVERSION, DEC → JAN

1.73% → 0.89%

A 49% relative drop in six weeks



TOTAL REVENUE (3 MO.)

\$362,165

Trending down into January

THE QUESTION THIS PROJECT ANSWERS

Is the decline about traffic, conversion, customer mix, or seasonality — and if it's conversion, exactly where in the journey are customers dropping off?

Five phases, each answering one business question

01



Data Exploration

Dataset:
bigquery-public-data.ga4_o
bfuscated_sample_ecomme
rce. Identifying events and
the schema.

02



Funnel Diagnostics

Understanding where we're
losing customers through
the funnel. Largest leak
found before checkout.

03



Segmentation

Paid traffic underperforms
free traffic every month; the
decline is seasonal, not
isolated to a segment.

04



Statistical Validation

Is the decline real? Isolated
to one segment?

05



Product-Level Analysis

Are specific products
underperforming?

What the funnel and the data actually show



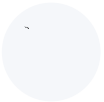
Conversion-driven, not traffic-driven

Sessions held flat (~2,300–4,600/day); conversion fell 1.73% → 0.89%, Dec to Jan.



The decline is statistically significant

Welch's t-test: $t = 4.332$, $p = 0.00006$ — not day-to-day noise.



Paid traffic underperforms every month

CPC converts lowest of all channels, Nov through Jan, without exception.



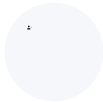
The biggest leak sits before checkout

Only 23% of sessions view a product; checkout-to-purchase is a healthy 42%.



It's seasonal, not a broken feature

Desktop and mobile conversion move together: $r = 0.81$.



Returning customers convert 5.3x better

6.67% conversion vs. 1.25% for new customers — the single largest lever.

Three tests: is it real, is it isolated, can it be trusted



Welch's t-test

Daily conversion is naturally noisy. Welch's variant doesn't assume equal variance, needed here since December's daily numbers swung harder than January's.

Is the Dec → Jan drop real, or normal daily noise?

$t = 4.332$
 $p = 0.00006$

SIGNIFICANCE

A gap this size would occur by chance only ~6 times in 100,000. Treated as a real behavioral shift — worth acting on, not noise.



Pearson Correlation

A device-specific bug (e.g. broken mobile checkout) would show up as one device tanking while the other holds steady. Correlation tests whether they move together.

Isolated to one device, or shared across both?

$r = 0.81$

SIGNIFICANCE

Desktop & mobile move together, day by day. Rules out a device-specific defect; supports a shared, seasonal cause.



OLS Linear Regression

Leadership wanted a revenue trend line. Before trusting a slope, it needed a gut-check: does one straight line fairly represent 3 months that include a holiday spike?

Can a simple trend be trusted for forecasting?

$\beta_1 = -44.27$ /day

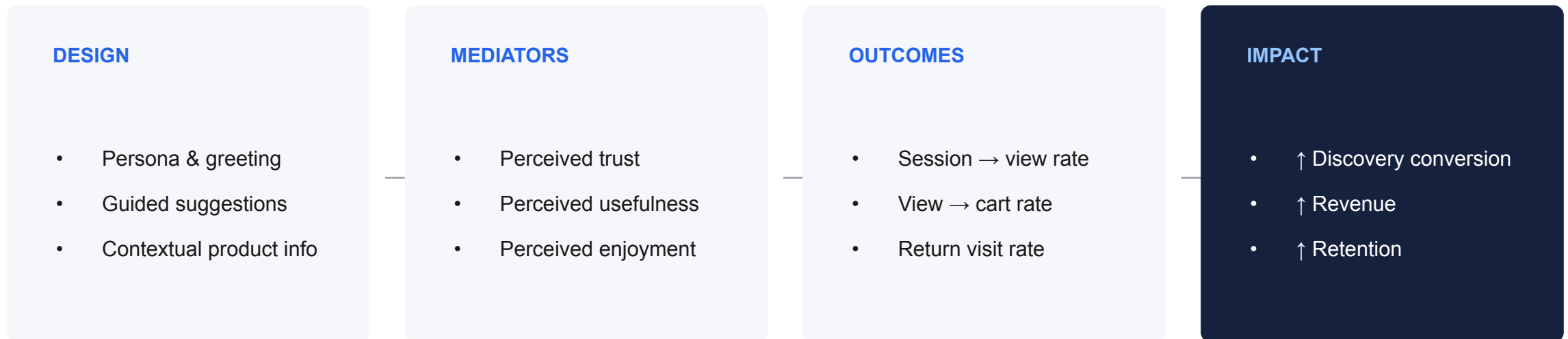
SIGNIFICANCE

No. The window includes the Nov/Dec holiday spike, so the slope overstates the decline — flagged, not used at face value.

RECOMMENDATION

Recommended product intervention: Nova

Analytics diagnosed a conversion-driven, pre-checkout leak (Finding #2). Behavioral research explains why: choice overload and low trust in an unfamiliar catalog. Nova, an AI discovery assistant, is the proposed fix for that specific gap — not a separate initiative.



Nova targets the exact gap Finding #2 identified: the pre-checkout drop-off. It doesn't discount — it reduces the friction of finding the right item in the first place.

Two segments, two very different conversion rates

Segments grounded directly in the segmentation data (Phase 3) — not assumed. Below: how the recommended intervention addresses each.



First-Time Browser

New visitor · 1.25% conversion

“There's a lot here. I don't know this brand yet — I'm not sure what's actually worth buying.”

NEEDS FROM NOVA

- Guided, low-effort discovery
- Social proof and context at first glance



Loyal Returner

Returning visitor · 6.67% conversion

“I already know what I like here. I just want to get back to it and check out.”

NEEDS FROM NOVA

- Personalized re-entry point
- Faster path to repeat and adjacent purchases

Rolling Nova out in three phases

A staged launch de-risks the investment: prove the discovery-conversion lift before scaling spend.



Phase 1 · Pilot

Months 1–2

- Launch Nova on the lowest-converting 20% of product pages
- A/B test against the existing browse experience
- Instrument session → view and view → cart tracking

Success gate: view → cart rate improves in the test group



Phase 2 · Expand

Months 3–4

- Roll out across the full catalog
- Add personalization: browsing history, geography
- Extend to returning-visitor re-entry flows

Success gate: discovery conversion lift holds at full scale



Phase 3 · Scale

Months 5–6+

- Extend Nova to email and push re-engagement
- Add mobile app surface
- Tie return-visit rate to lifetime value reporting

Success gate: measurable lift in return visit rate & retention

What closing the conversion gap is worth

Modeled off the observed conversion gap, not a promised outcome — three scenarios based on how much of the Dec → Jan gap Nova recovers.

CONSERVATIVE

25% of gap closed

\$152,500

projected annualized revenue

\$12,700 / month

MODERATE

50% of gap closed

\$305,200

projected annualized revenue

\$25,400 / month

STRONG

75% of gap closed

\$457,800

projected annualized revenue

\$38,100 / month

Assumptions: ~89,000 sessions/month (267,116 ÷ 3), the observed 0.84-point Dec → Jan conversion gap (1.73% → 0.89%), and an average order value of ~\$68. These are directional planning estimates, not measured results — the Phase 1 pilot is designed to validate the real recovery rate.



SUMMARY

The analysis pointed to one recommendation



The decline was conversion-driven and statistically confirmed — not noise, not a device bug.



Nova is the recommended intervention for the exact pre-checkout leak the funnel data identified.



A phased rollout puts \$150K–\$460K in projected annual revenue on the table.

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